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# What explains the performance of community-based initiatives? Testing the impact of leadership, social capital, organizational capacity, and government support

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## ABSTRACT

Community-based initiatives (CBIs) are thriving in Western countries. In CBIs, citizens take a leading role in providing public services and goods. CBIs have been acclaimed for their innovativeness, problem-solving capacity, and legitimacy. However, we lack large N studies on performance of CBIs and its antecedents. This article develops and tests a model that identifies relationships between performance and four antecedents by using survey data on CBIs collected in the Netherlands (N = 671). Using structural equation modelling, positive direct and indirect relationships between transformational leadership, boundary spanning leadership, organizational capacity, social capital ties, government support, and performance are found.

**KEYWORDS** Community-based initiatives; performance; community leadership; government support; social capital

## Introduction

The idea of public administration shifting into a new paradigm of co-creation is growing across public sector scholars and practitioners (cf. Torfing, Sørensen, and Røiseland 2019; Voorberg, Bekkers, and Tummers 2015). This paradigm, also known as New Public Governance, identifies citizens and other service-users as part of the production process, in which citizens are given more responsibility in the creation of public services (cf. Brandsen, Trommel, and Verschuere 2017; Osborne, Radnor, and Strokosch 2016). A specific phenomenon within this current co-creation and co-production discourse, are community-based initiatives (CBIs) aimed at the self-organization of public services.

In CBIs, instead of co-creating or co-producing under conditions and frameworks set by governments, citizens take the lead and collectively initiate and implement initiatives aimed at providing public goods or services for their community. CBIs are emerging in Western countries, apart from the established and institutionalized civil society (Brandsen, Trommel, and Verschuere 2017). These initiatives have been

acclaimed for their innovativeness, problem-solving capacity, and legitimacy (Edelenbos and Van Meerkerk 2016; Torfing, Sørensen, and Røiseland 2019).

However, there are also doubts about their scale and impact (Brandsen, Trommel, and Verschuere 2017). In general, we know little about the performance of these CBIs and the factors that explain their performance. We lack empirical research and especially large N studies on the antecedents that explain the performance of CBIs (Igalla, Edelenbos, and Van Meerkerk *forthcoming*; Torfing, Sørensen, and Røiseland 2019). Existing research on CBIs is strongly dominated by case study research, lacking large-scale testing of theoretical relationships.

In this article, we develop and test a conceptual model of CBIs that identifies relationships between (a) factors regarding community initiatives and (b) the performance of said initiatives by using survey data collected in the Netherlands (N = 671 people participating in CBIs). Four key factors derived from different literature fields and identified as key factors in a systematic literature review on CBIs (Igalla, Edelenbos, and Van Meerkerk *forthcoming*) are: leadership styles of citizens leading the initiatives (cf. Purdue 2001); the organizational capacity of initiatives (cf. Bailey 2012); support of government (cf. Seixas and Berkes 2009), and social capital ties (cf. Newman et al. 2008).

This article seeks to contribute to the literature on CBIs by testing a theoretical model on key antecedents of performance. The literature considers each of the factors as influential on performance of CBIs, but they have not been combined into a comprehensive model and systematically tested for their actual relationships with performance. Being strongly embedded in theory, our results have also relevance for non-Dutch CBIs. We formulate the following main research question:

*How do key antecedents influence the performance of community-based initiatives?*

## Theoretical framework

### *Defining community-based initiatives and their performance*

We define CBIs as a form of self-organization in which citizens mobilize resources to collectively define and carry out projects aimed at providing public goods or services for their community (see also Igalla, Edelenbos, and Van Meerkerk *forthcoming*). Citizens control the aims, means, and actual implementation of their activities (Healey 2015), which resembles the highest level of Arnstein's (1969) participation ladder.

Although CBIs are in charge of internal matters (e.g. policy and managerial aspects), they are often linked to government and various other actors, such as funding organizations and traditional third sector organizations (cf. Healey 2015). CBIs often operate in institutionalized settings with regulations at multiple scales and therefore interact with government (e.g. Edelenbos and Van Meerkerk 2016; Healey 2015). This precise characteristic of CBIs does not implicate an absence of government or other helping hands, but indicates citizens being in control of the provision of public services, leading a hybrid network of support through bonding, bridging and linking ties (cf. Voorberg, Bekkers, and Tummers 2015). This makes CBIs different from regular co-production, because citizens are not 'just' involved in producing public services under frameworks set by government, but they initiate

and lead whereas governments follow and facilitate (cf. Alford, 2002; Arnstein, 1969; Boonstra and Boelens 2011). This feature makes CBIs also distinct from dialogue and deliberation approaches, such as citizen juries (Nabatchi 2012; Roberts 2004). This focus on service provision instead of policy or politics makes them also different from social movements in which a diverse range of activists (both individuals and organizations) mobilize themselves to achieve collective political goals (cf. Nicholls 2009).

In addition, CBIs can differ in their level of formalization; we include both formalized and informal initiatives. When formalized, differences can be found in their legal structure (e.g. cooperative, community enterprise, etc.), but in this article all CBIs share the same characteristics found in a systematic literature review on CBIs (Igalla, Edelenbos, and Van Meerkerk *forthcoming*): a (formal/informal) form of self-organization, providing public services or goods to a community, being in control of internal decision-making, not-for-private-profitmaking, mainly operating on voluntary work, and being community-based (cf. Bailey 2012; Llano-Arias 2015). These characteristics sets CBIs apart from activities of professionalized non-profit organizations in the traditional third sector with paid workers and no link to voluntary citizen participation (cf. Edelenbos and Van Meerkerk 2016). Because our aim in this article is to explain the performance of CBIs, rather than for instance understand how and why citizens organize themselves to create collective goods, how individual and community interests can be balanced, and how their efforts become institutionalized (cf. Douglas 1991; Ostrom 2000), we focus on theories concerning resource mobilization and coordination, with antecedents such as social capital and leadership, instead of approaching CBIs from a collective action perspective and Ostrom's work on common-pool resources.

### ***Performance measurement: a multi-categorical perspective***

The concept of performance is complex and multi-interpretable, especially in the public (and voluntary) sector (cf. Andrews, Boyne, and Walker 2011; Kendall and Knapp 2000) in which CBIs are active. It is insufficient to evaluate success or failure using financial criteria, as oppositional to the private sector, performance in the public and non-profit sector is multidimensional and related to a great variety of stakeholders (e.g. government, funders, volunteers, board members, service users) (Andrews, Boyne, and Walker 2011; Kendall and Knapp 2000). Which performance categories and criteria to select can, therefore, vary among the different interests and expectations of stakeholders. There is no simple and uncontested way to measure performance of CBIs (cf. Kendall and Knapp 2000). Instead, it is important to recognize that performance of CBIs, as is the case for public sector and voluntary organizations, comprises different categories or dimensions (cf. Andrews, Boyne, and Walker 2011; Kendall and Knapp 2000). The literature on public management shows different important dimensions to consider, including efficiency, effectiveness, quality of outputs, responsiveness to service needs, and innovation (e.g. Boyne 2002). Likewise, literature on social enterprises and networks discuss similar dimensions of performance (cf. Klijn, Edelenbos, and Steijn 2010; Liu, Takeda, and Ko 2014) and especially stresses the importance of both economic and social categories for non-profits (cf. Bagnoli and Megali 2011; Liu, Takeda, and Ko 2014). Indeed, considering the characteristics of CBIs, it is important to not only focus on organizational performance, but also on community performance, as CBIs aim to create broader public value (cf. Bagnoli and Megali 2011; Igalla, Edelenbos, and Van Meerkerk

forthcoming). Therefore, we define performance of CBIs as the multi-categorical achievement of community and organizational outcomes resulting from self-organizing one or more specific public services and/or goods. In the methodology section, we further discuss the specific conceptualization and measurement of performance.

### ***Organizational capacity: importance of financial and human resources***

Organizational capacity is about the ability of an organization to fulfil its mission (Eisinger 2002:115, 117). It distinguishes common features of organizations that enable the production of desired outcomes (cf. Foster-Fishman et al. 2001). In this article, we focus on two important features for CBIs: financial and human resources (e.g. Foster-Fishman et al. 2001). Human resources concern volunteers participating in CBIs and they form an important work force as CBIs often (solely) operate on a voluntary basis (e.g. Bailey 2012; Healey 2015). Committed volunteers provide resources of time and energy that increase the capacity of initiatives to achieve the desired outcomes (e.g. Foster-Fishman et al. 2001; Healey 2015).

Next, CBIs need financial resources for various ends, including initiating and running services, to implement new activities, pay for buildings, invest in communication and exposure, and mobilize volunteers (e.g. Foster-Fishman et al. 2001; Healey 2015). CBIs can have multiple revenue sources, which is argued to be positively associated with their success and growth (e.g. Sharir and Lerner 2006). Donations, finances from charitable funders, sponsors, and the private sector are also common. Initiatives are often active in generating earned income, for instance through registration fees and selling of products (e.g. Bailey 2012).

Based on the above discussion, we hypothesize that:

**Hypothesis 1a.1:** Organizational capacity, in terms of volunteers, is positively associated with the performance CBIs.

**Hypothesis 1a.2:** Organizational capacity, in terms of revenue sources, is positively associated with the performance of CBIs.

### ***The role of government support***

Even though CBIs operate with a high level of control, and might occur because citizens feel that governments are not performing sufficiently and therefore want to organize certain services themselves, they are often linked to government institutions (Edelenbos and Van Meerkerk 2016). As CBIs operate in the public domain and do not seldom interfere with institutionalized processes of service delivery or policy and decision-making, they are often dependent on how local government respond to their initiative (Brandsen, Trommel, and Verschuere 2017). To develop effective and successful collective actions over time, citizens need at least minimal recognition of the right to organize by government (Ostrom 2000). Moreover, government support of their initiatives is useful for getting started or for gaining assets (Bailey 2012). In their comparative case study on 10 ‘successful’ CBIs in different South-American

countries, Seixas and Berkes (2009) found that these CBIs had most of the times supportive relationships with government organizations.

Governments can have different reasons to support CBIs. Especially in light of budget cutbacks and current administrative philosophy that stresses co-creation and citizen engagement (e.g. Torfing, Sørensen, and Roiseland 2019), CBIs can be of strategic interest, for instance to maintain a certain level of service delivery in the fields of retreat or to achieve policy goals (e.g. Edelenbos and Van Meerkerk 2016). Governments can provide a range of services and support functions for CBIs, including start-up funds, business networking and marketing, technical training and knowledge transfer (Korosec and Berman 2006; Healey 2015; Edelenbos, Van Meerkerk, and Schenk 2018). As government support has been claimed to be crucial for the success of CBIs (cf. Dale and Newman 2010; Healey 2015), we formulate the following hypothesis:

**Hypothesis 2:** Government support is positively associated with the performance of CBIs.

### ***Social capital: bonding, bridging, and linking ties of community initiatives***

Theories on social capital are highly relevant for explaining the functioning and performance of CBIs. Social capital is defined by Putnam (1995, 664–665) as 'features of social life – networks, norms, and trust – that enable participants to act together more effectively to pursue shared objectives'. Social capital facilitates the mobilization of resources and the coordination of action. CBIs heavily rely on social capital, because their access to economic capital as volunteer organizations is usually limited (Newman et al. 2008). We use the common distinction between bonding, bridging, and linking social capital. Bonding social capital refers to 'trusting and co-operative relations between members of a network who see themselves as being similar, in terms of their shared social identity' (Szreter and Woolcock 2004, 654–655). In the context of CBIs, a shared social identity is present among members of the core group, the driving force of the initiatives (cf. Newman et al. 2008). Bridging social capital refers to relations of exchange, respect and mutuality between people who see themselves to be unlike in some social identity sense, and linking social capital refers to ties of exchange between actors who know themselves to be unequal in their power and access to resources (Szreter 2002, 579). In the context of CBIs, bridging ties can refer to ties connecting target groups or other associations operating in the community. Linking ties are often present through connections with (local) government (agencies) and other institutions, like funding agencies (cf. Dale and Newman 2010; Szreter 2002).

Literature shows indirect influence of social capital on performance of CBIs through leadership and organizational capacity. Specifically, community organizing requires strong bonding ties (cf. Dietz, Ostrom, and Stern 2003), which can usually be found in the core group of CBIs (cf. Dale and Newman 2010). This core group functions as backbone of the CBI (Dale and Newman 2010), committed to organize activities. Strong bonding ties in the core group help to increase revenue sources, facilitate communication within initiatives and mobilize volunteers who want to pursue the same goals (cf. Varda 2011). We, therefore, hypothesize that:

**Hypothesis 3a.1:** Bonding social capital is positively associated with organizational capacity (human resources) of CBIs.

**Hypothesis 3a.2:** Bonding social capital is positively associated with organizational capacity (revenue sources) of CBIs.

Building on the literature of linking and bridging capital, we argue that a greater number of linking ties helps initiatives to prove that they are able to connect with institutional actors, showing their legitimacy to government authority. Moreover, it may increase the credibility of CBIs as providers of public services and may lead to greater commitment of government representatives to invest public resources in the CBIs. Furthermore, bridging ties enhance the organizational capacity of CBIs by mobilizing resources from the community, as ties with for instance other community organizations, and residents provide a pool of potential volunteers and other resources, such as materials, and financial contributions (e.g. Bailey 2012). Based on these findings and arguments, we form the following three hypotheses:

**Hypothesis 3b:** Linking social capital is positively associated with government support obtained by CBIs.

**Hypothesis 3c.1:** Bridging social capital is positively associated with organizational capacity (human resources) of CBIs.

**Hypothesis 3c.2:** Bridging social capital is positively associated with organizational capacity (revenue sources) of CBIs.

### ***Leadership styles: leading the organization and managing its external network***

Leadership can be defined as “mobilizing people to tackle tough problems” (Heifetz, as quoted in: Hartley and Allison 2000, 36). This definition approaches (community) leadership as a set of dynamics occurring among and between individuals, groups and organizations. The motivation of people, and the achievement of outcomes is underlined (Hartley and Allison 2000). Community leaders act on two levels; on the organizational level they lead operational, strategic, and relational aspects of initiatives, and on the community level they are active in mobilizing, creating, and maintaining ties to external actors (cf. Purdue 2001).

Regarding community leadership, we will focus on transformational leadership as intra-organizational style and on boundary spanning leadership as inter-organizational leadership style.

### ***Transformational leadership as intra-organizational style to lead CBIs***

Transformational leadership (TFL) is based on directing and inspiring followers “by raising their awareness of the importance of organizational values and outcomes” (Wright, Moynihan, and Pandey 2012, 207). Transformational leaders are able to articulate a clear and inspirational agenda of change, expressing an appealing vision of the organization’s mission and future (Phillips and Pittman 2009). In addition, transformational leaders stimulate and encourage creativity and innovativeness of those around them (Bass et al. 2003).



As Wright et al (2012, 207) point out; TFL can be particularly useful in non-profit organizations because of their strong service- and community-oriented missions. Moreover, in the context of CBIs, services are being self-organized in order to pursue higher-order societal goals. So, volunteers can be expected to be intrinsically motivated and wanting something more than just being extrinsically rewarded for their contribution (c.f. Alford 2002), which fits with the principles of TFL. To illustrate this point, Alford (2002) distinguished three intrinsic motivations to become engaged in the community: increasing your knowledge of the world and developing your skills, expressing altruistic concerns, and developing psychologically and enhance your esteem. Such motivations can be expressed in environments that allow people to develop their selves, to be inspired to pursue ambitious goals, and to be entrepreneurial, thinking of new ideas for the community. Such an environment characterizes CBIs, which are known for their strong focus on the community and social relationships, and their drive to develop their own services, with room for experimenting (cf. Boonstra and Boelens 2011; Voorberg, Bekkers, and Tummers 2015). Transformational leadership can be seen as leadership style that enhances such an environment, for instance by its ability to become a source of inspiration for the volunteers, and the ability to foster intellectual stimulation, two important dimensions of TFL (see also methodology for the exact measurement). Various studies in different settings (e.g. corporate and public) have shown positive relationships between TFL and performance (see Hater and Bass 1988; Wright, Moynihan, and Pandey 2012). Although TFL has been labelled as effective and crucial leadership style for performance of CBIs, less is known about its actual contribution in this context (e.g. Phillips and Pittman 2009). However, we expect to find the same strong and direct influence of TFL on performance in the context of CBIs:

**Hypothesis 4:** Transformational leadership is positively associated with the performance of CBIs.

Foster-Fishman et al. (2001) indicate that leadership is an important factor to build organizational capacity. In order to mobilize and retain resources, both human and revenue resources, it is important for community leaders to articulate an inspiring vision and agenda that attracts people and organizations to invest their time, energy, and financial resources to achieve the collective goals. Community leaders adopting TFL can inspire others, communicate a clear vision, and provide long-term plans (e.g. Wright, Moynihan, and Pandey 2012). These different characteristics of TFL help with mobilizing resources, and commitment from residents and organizations to build capacity and achieve common goals. We, therefore, expect a positive relationship between TFL and the organizational capacity of CBIs:

**Hypothesis 5a.1:** Transformational leadership is positively associated with the organizational capacity (human resources) of CBIs.

**Hypothesis 5a.2:** Transformational leadership is positively associated with the organizational capacity (revenue resources) of CBIs.



In relation to social capital, community leaders affect the level of social capital with their leadership styles. As Purdue (2001, 2214) states: “community leaders are engaged in accumulating internal social capital embodied in their accountability to their grassroots following and external social capital in their access to wider ... networks”. Transformational leadership focuses on developing an organizational vision and future. This vision, in turn, helps to develop common ground and orientation between people, making them feel more connected with each other, which strengthens bonding ties. This argument leads to the following hypothesis:

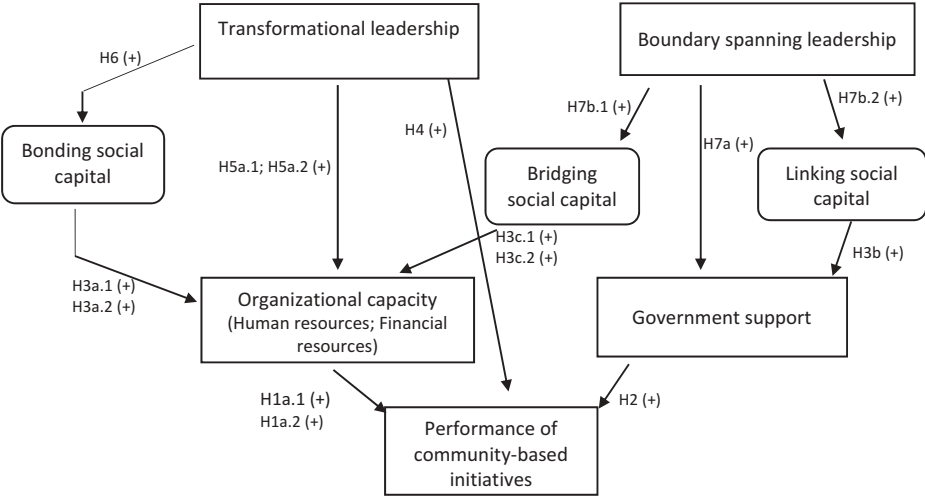
**Hypothesis 6:** Transformational leadership is positively associated with bonding social capital of CBIs.

### *Boundary spanning leadership to lead networks of CBIs*

Another strand of literature pays attention to the external orientation and activity of leaders. Boundary spanning leadership (BSL) stresses the importance for organizations to survive and to enhance their performance by adapting to the environment and creating a better fit (Aldrich and Herker 1977). In particular, in a context of interdependencies and scarce resources, boundary spanning activities are considered to be important for gaining necessary resources and linking the organization to external developments which might create opportunities for innovation and growth of the organization (Van Meerkerk and Edelenbos 2018). Different types of boundary spanning activities are accentuated in the literature: linking to potential partners and building sustainable relationships, managing information flows, and connecting to relevant external developments and processes (Van Meerkerk and Edelenbos 2018). In this respect, different types of competences are stressed, such as having a good feeling for the interest of other actors (otherness), empathy, communicative capacity and conflict management skills (Williams 2002). Boundary spanning activities and competences are important in developing the external orientation and network of (community) organizations. As many CBIs are dependent on acquiring external resources and support, especially from government, often boundary spanning activities are focused at acquiring this governmental support (Edelenbos, Van Meerkerk, and Schenk 2018). Boundary spanners are organizers as well as institutional infiltrators (Miller 2008) as they know how to enter governmental institutions and find their path to people at positions who can help them. Community boundary spanners become important in navigating the initiative through the governmental system, arriving at the right departments and people to generate administrative and political support for the community initiative. From this line of reasoning, we arrive at the following hypothesis:

**Hypothesis 7a:** Boundary spanning leadership is positively associated with government support of CBIs.

In relation to social capital, community leaders affect the level of social capital with their leadership styles. Boundary spanning leadership is important to develop and maintain new relationships (Miller 2008), connecting different community members or institutions, thereby specifically oriented at creating linking and bridging capital



**Figure 1.** Conceptual model to influence the performance of community-based initiatives. Note: (+) = positive relationship.

(cf. Dale and Newman 2010). For CBIs, having a sense for local issues, actively maintaining contact with residents and other CBIs in the community, and involving them with the initiative, can strengthen bridging links in the community (Igalla, Edelenbos, and Van Meerkerk [forthcoming](#)). Likewise, devoting time to maintain contact with institutional actors, and having knowledge of what is important for them (cf. Van Meerkerk and Edelenbos 2018), can help CBIs in connecting their goals with policy, needs, and agendas of institutions, thereby strengthening linking ties (Igalla, Edelenbos, and Van Meerkerk [forthcoming](#)).

We therefore hypothesize that:

**Hypothesis 7b.1:** Boundary spanning leadership is positively associated with bridging social capital of CBIs.

**Hypothesis 7b.2:** Boundary spanning leadership is positively associated with linking social capital of CBIs.

Above we have provided a theoretical argumentation for several associations between variables. In [Figure 1](#), which depicts our conceptual framework, we summarize all the hypothesized associations.

## Research methods

### Data collection and sample

We conducted a web-based survey among Dutch respondents participating in CBIs and collected the data in November 2016 (from the 8<sup>th</sup> until the 16<sup>th</sup> of November, including one reminder on November the 14<sup>th</sup>).

A particular difficulty of the target population is that inclusive lists of people participating in CBIs do not exist. Therefore, the sample was drawn from online panels

managed by Kantar Public. Kantar Public is an integrated consulting and research agency that works across the world and aims to improve public policy, public services and public communication. For the Netherlands, this organization has a large reach of their online panel and representativeness, which consists of 124,000 randomly selected respondents from different social-economic categories. We included a screening to identify respondents participating in community-based initiatives, aiming to rule out respondents active in other kinds of civic participation, such as formal political participation, and activism (e.g. writing petitions). People participating in the online panels were asked whether they are or were involved (in the last year) in a citizen initiative, described as “activities that are organized by people *themselves* with little or no involvement from the government”. We provided some examples, like grassroots activities focused on providing care facilities, starting community enterprises, maintaining children’s playgrounds, and caring for public spaces and planting greenery. If they did, then they were invited to complete the full questionnaire. From this screening, 1500 people were representatively (age, social class, education, gender, etc.) selected to complete the full questionnaire. Of these respondents, a group of 797 respondents actually participated in the research (response rate of 53, 13%). After screening the data to ensure the quality of response and to ensure the respondents were indeed participating in CBIs (e.g. by checking the description of the initiative and available information on the web), a total of 671 respondents remained in the dataset (adjusted response rate of 44, 73%). Specific examples of CBIs in our sample include for instance community associations aimed at youth, women or people with disabilities; neighbourhood watches; CBIs maintaining greenery; energy cooperatives, and initiatives focused on providing care for the elderly.

Of the 671 respondents, 54% were male, the average age was 56.45 years ( $SD = 14.10$ ), the predominant educational level is higher vocational education (31.89%), and the most common status of employment is paid employment (40.83%). Considering the role of the respondents (see also control variables), most respondents were active volunteers (60.95%), followed by board members (25.63%), and passive or supporting volunteers (11.92%) (and other (1.49%)). Thus, there is no dominance of board members in our sample (at least 72.88% of the respondents is not a board member), which reduces the risk of overly positive self-reporting for concepts like leadership. We included the role of the respondent as a control; no significant relationships were found concerning our dependent variable.

## **Measurement of variables**

### **Performance**

Quantitative research on performance of CBIs is scarce (Igalla, Edelenbos, and Van Meerkerk [forthcoming](#)). A proven scale for measuring their performance is lacking, so we developed our own scale. Our scale consists of multiple-criteria, which takes the multi-categorical feature of performance of CBIs into account as explained in the theoretical section. For the development of the criteria, we build upon existing scales in literature on public sector, network, and social enterprise performance, and made them context-specific with the use of (qualitative case) studies about outcomes of CBIs (e.g. Bagnoli and Megali 2011; Klijn, Edelenbos, and Steijn 2010). The criteria are based on seven broad outcomes: goal effectiveness, problem-solving capacity, efficiency, innovativeness, quality of services, social impact on community, and

legitimacy, which together measure performance as a first-order construct. The specific items are shown in Table 1 and they form together a one-component structure.<sup>1</sup> We used a seven-point Likert scale, with 1 “absolutely not” (agreeing with the statement presented in the item) and 7 “very strongly” (agreeing with the specific statement).

We measure performance through individual respondents’ perceptions of these organizational and community level outcomes. Furthermore, we used perceived performance as a proxy for objective performance, which is also common in the field of third sector organizations and in the general literature on network performance (cf. Klijn, Edelenbos, and Steijn 2010; Liu, Takeda, and Ko 2014). However, this type of measure has also its downside as it means self-reporting data instead of (or in combination with) more tangible measures for performance (see more on this limitation in our conclusions).

### *Leadership styles*

We used previously tested scales to measure TFL (e.g. Hater and Bass 1988; Wright, Moynihan, and Pandey 2012), and BSL (Van Meerkerk and Edelenbos 2018) as multi-dimensional concepts by adopting the important dimensions of both leadership styles from the literature. The items of each scale were made context-specific for CBIs (see Table 1 for the items). All items of the leadership styles were measured using a seven-point Likert scale (from (1) “absolutely not” to (7) “very strongly”).

### *Government support*

We measured the level of perceived government support by building on the scales of Korosec and Berman (2006). They distinguish three types of support that are highly interrelated and can therefore function together as a measure for government support. In this article, we used several of their items and made them context-specific for CBIs, resulting in 11 items (see Table 1). We used a seven-point Likert scale to measure the items, ranging from (1) “absolutely not” to (7) “very strongly”.

### *Social capital*

To measure bonding social capital, we used the frequency of the interactions among members in the core group (to represent the strength of strong ties), which is common in research about social capital (cf. Varda 2011). We used a six-point scale with the following scores: (0) never, (1) once per year, (2) every few months, (3) every few weeks, (4) weekly, and (5) daily and recoded the option ‘I do not know’ into missings.

For bridging and linking social capital we asked respondents to score the frequency of interaction (to represent the strength of weak ties) with the bridging and linking actors that are often linked to (Dutch) CBIs (see Table 1 for the actors). We used a seven-point scale to measure the frequency of interaction with each actor using the following scores: (0) never, (1) less than once per year, (2) once per year, (3) every few months, (4) every few weeks, (5) weekly, and (6) daily. Thereafter, we made two index scores, respectively, summing up frequencies of bridging and linking actors.

**Table 1.** Measurement items and construct's reliability.

| Constructs and items                                                                                         | Standardized factor loadings | AVE (convergent validity) & Squared correlation (discriminant validity) | Cronbach's Alpha/ Composite reliability |
|--------------------------------------------------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------|-----------------------------------------|
| <b>Community-based initiative performance</b><br>The citizen initiative in which I am (was) involved ...     |                              |                                                                         | 0.85/0.86                               |
| 1) achieves its objectives                                                                                   | .78                          | 0.46                                                                    |                                         |
| 2) provides answers to key issues                                                                            | .63                          | Performance – TFL: 0.38                                                 |                                         |
| 3) creates better connections between residents/citizens (social cohesion)                                   | .66                          | Performance – BSL: 0.21                                                 |                                         |
| 4) manages to deliver high-quality services and goods                                                        | .73                          |                                                                         |                                         |
| 5) is considered important by the community                                                                  | .74                          | Performance – GS: 0.12                                                  |                                         |
| 6) finds smarter solutions to problems than previous attempts                                                | .65                          |                                                                         |                                         |
| 7) earns a sufficient income to cover expenses                                                               | .53                          |                                                                         |                                         |
| <b>Transformational leadership (TFL)</b><br>The management (board or leading persons) of the initiative ...  |                              | 0.72                                                                    | .93/.93                                 |
| 1) has a clear idea (vision) of where the organization is heading                                            | .84                          | TFL – BSL: 0.53                                                         |                                         |
| 2) is a source of inspiration for the members/volunteers within its organization                             | .89                          | TFL – GS: 0.10                                                          |                                         |
| 3) makes plans that display visible leadership to its members/volunteers and followers                       | .86                          |                                                                         |                                         |
| 4) develops long-term plans for the organization                                                             | .78                          |                                                                         |                                         |
| 5) challenges me to solve existing problems in new ways                                                      | .86                          |                                                                         |                                         |
| <b>Boundary spanning leadership (BSL)</b><br>The management (board or leading persons) of the initiative ... |                              | 0.71                                                                    | .93/.93                                 |
| 1) involves people from outside the organization when making decisions                                       | .78                          | BSL – GS: 0.21                                                          |                                         |
| 2) works closely with others (outside the organization) to achieve results                                   | .87                          |                                                                         |                                         |
| 3) aims to link external developments (new opportunities, possibilities, etc.) to the citizen initiative     | .87                          |                                                                         |                                         |
| 4) devotes a lot of time to maintaining contact with parties outside the organization                        | .86                          |                                                                         |                                         |
| 5) has a sense of what is important for parties outside the initiative                                       | .84                          |                                                                         |                                         |

(Continued)

Table 1. (Continued).

| Constructs and items                                                                                                             | Standardized factor loadings | AVE (convergent validity) & Squared correlation (discriminant validity) | Cronbach's Alpha/ Composite reliability |
|----------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------|-----------------------------------------|
| <b>Government support (GS)</b>                                                                                                   |                              |                                                                         |                                         |
| The municipality ...                                                                                                             |                              | 0.66                                                                    | .95/.95                                 |
| 1) provides financial assistance to the initiative (such as subsidies)                                                           | .67                          |                                                                         |                                         |
| 2) supports the initiative in obtaining extra resources (e.g. acquisition, submitting applications for subsidies, fundraising)   | .80                          |                                                                         |                                         |
| 3) helps the initiative by providing availability to real estate (buildings) or land                                             | .63                          |                                                                         |                                         |
| 4) gives the initiative the opportunity to execute assignments (e.g. waste collection, maintenance of public green spaces, etc.) | .75                          |                                                                         |                                         |
| 5) supports the initiative through the provision of information                                                                  | .88                          |                                                                         |                                         |
| 6) contributes towards initiative awareness                                                                                      | .88                          |                                                                         |                                         |
| 7) assists in the coordination with other involved parties                                                                       | .88                          |                                                                         |                                         |
| 8) encourages collaboration between those parties with an interest in the initiative                                             | .85                          |                                                                         |                                         |
| 9) provides the initiative with advice, if required                                                                              | .86                          |                                                                         |                                         |
| 10) actively cooperates with the initiative                                                                                      | .89                          |                                                                         |                                         |
| 11) participates in the initiative by taking responsibility for certain tasks                                                    | .80                          |                                                                         |                                         |
| <b>Organizational capacity</b>                                                                                                   |                              |                                                                         |                                         |
| Revenue sources (N = 616)                                                                                                        | N/A                          | N/A                                                                     | N/A                                     |
| How does the citizen initiative earn an income? Multiple answers are possible:                                                   |                              |                                                                         |                                         |
| 1) subsidies from the local government (e.g. the municipality) (yes/no)                                                          |                              |                                                                         |                                         |
| 2) subsidies from the regional government (e.g. county/province or city region) (yes/no)                                         |                              |                                                                         |                                         |
| 3) subsidies from the Dutch government (e.g. ministries) (yes/no)                                                                |                              |                                                                         |                                         |
| 4) services to the government (e.g. via a contract or social contract) (yes/no)                                                  |                              |                                                                         |                                         |
| 5) private revenue (the sale of goods and services to a non-governmental party) (yes/no)                                         |                              |                                                                         |                                         |
| 6) donations (yes/no)                                                                                                            |                              |                                                                         |                                         |
| 7) fundraising (e.g. by a foundation or bank) (yes/no)                                                                           |                              |                                                                         |                                         |
| 8) contributions (from members/volunteers) (yes/no)                                                                              |                              |                                                                         |                                         |
| Human resources                                                                                                                  |                              |                                                                         |                                         |
| How many people (volunteers and/or active members) are active in the citizen initiative (scale 1–6)?                             |                              |                                                                         |                                         |

(Continued)

Table 1. (Continued).

| Constructs and items                                                                                                                                                                                                                     | Standardized factor loadings | AVE (convergent validity) & Squared correlation (discriminant validity) | Cronbach's Alpha/ Composite reliability |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-------------------------------------------------------------------------|-----------------------------------------|
| <b>Social capital</b>                                                                                                                                                                                                                    |                              |                                                                         |                                         |
| Bonding social capital (N = 577)                                                                                                                                                                                                         |                              |                                                                         |                                         |
| - How frequently do members of the core group interact with issues that affect the citizen initiative? This interaction can be in various forms, such as attending management meetings, organizing activities and/or providing services. | N/A                          | N/A                                                                     | N/A                                     |
| Bridging and linking social capital (N = 670)                                                                                                                                                                                            |                              |                                                                         |                                         |
| Generally speaking, how often is the citizen initiative in contact with the following parties? This contact can take various forms, e.g. working together, performing a financial consultation and/or receiving advice.                  |                              |                                                                         |                                         |
| <i>Bridging social capital:</i><br>Residents organizations/other community initiatives, and visitors or users of the initiative                                                                                                          |                              |                                                                         |                                         |
| <i>Linking social capital:</i><br>Central government, province (county), municipality: local council, municipality: municipal executive, municipality: civil servants, funds/sponsors, housing corporations                              |                              |                                                                         |                                         |

If not mentioned otherwise in the table, N = 671. All factor loadings:  $p < .001$



### **Organizational capacity**

We focus on two important organizational features of CBIs, human resources and revenue sources, as a proxy to measure their organizational capacity (e.g. Foster-Fishman et al. 2001). To measure human resources, we asked respondents about the average number of volunteers and/or active members participating in the CBI, using a six-point scale ranging from 1 to 10 volunteers (1) to 81 or more volunteers (6). For revenue sources, we asked about the presence of different revenue sources in the CBI (see Table 1 for the items) and computed an index by summing all items, which has a range between 0 (no revenue sources) and 6 (six different revenue sources) in our data.

### **Data analysis**

In order to test our hypotheses, we use structural equation modelling (SEM) with Amos version 24. To carry out the SEM analyses, we follow the two-step modelling approach introduced by Anderson and Gerben (1988), creating a measurement and a structural model. The measurement model was modified when necessary. One modification to enhance the model included correlations of errors,<sup>2</sup> which can be used if “in case of multiple questionnaire items, correlated errors may arise from items that are very similarly worded...” (Brown 2015, 157). This seems to be an important reason in our research (see footnote 2 for the items for which we have correlated the errors, and Table 1 for the wording of the items). Furthermore, because we developed a new scale, we randomly split our sample into two subsamples to validate the scale and used the main sample ( $N = 671$ ) for the measurement and structural equation model. See Appendix 1 for further information on the scale validation process.

### **Reliability and validity**

The measurement model has been examined for convergent and discriminant validity based on the confirmatory factor analyses. For the sample, all factor loadings are  $\geq .53$  and most go beyond .70, which is a first indicator to demonstrate convergent validity (Hair et al. 2010). A second indicator is the AVE, which is for most variables higher than the threshold of .50, except for performance (see Table 1). However, comparing the shared variance (squared correlation) between each pair of constructs against the AVEs for these two constructs, we found evidence for discriminant validity for all constructs. In addition, the composite reliability values of the constructs are high and exceed the .70 threshold. Furthermore, all constructs have Cronbach’s Alpha’s greater than the generally accepted value of .80.

### **Common method bias**

We need to address common method bias (CMB) as our study was based on self-reported data obtained from the same source (Podsakoff et al. 2003). In this study we focus on perceived performance and not on objective performance, which is common in the field of third sector organizations and in the general literature on network performance (e.g. Klijn, Edelenbos, and Steijn 2010; Liu, Takeda, and Ko 2014), making self-reports a relevant measurement method (cf. Conway and Lance 2010). Moreover, assessing objective performance in large  $N$  research is difficult or even impossible to achieve. In addition, other data sources that measure performance of

CBIs (such as archival data) are unavailable, which makes the use of the survey as single method plausible (cf. George and Pandey 2017). Next, for most variables (including performance) we used multiple items in our measurement, and the scale reliability of all relevant constructs are well above general thresholds. Both reduce the likelihood of CMB (cf. George and Pandey 2017).

We used ex-ante procedural remedies for reducing the likelihood of CMB in the survey design and ex-post statistical controls for testing CMB. Procedurally, we reduced the risk of CMB by allowing the respondents' answers to be anonymous (Podsakoff et al. 2003, 888) and we separated between measures in the survey. In addition, we carefully constructed the items, by pretesting the survey among researchers and practitioners. This helps in reducing item ambiguity (Podsakoff et al. 2003).

Statistically, we conducted the Harman one-factor test in SPSS and the unmeasured latent method construct in AMOS to assess whether the majority of the variance could be explained by a single factor (Podsakoff et al. 2003). Results confirmed that CMB is not a major concern in this study.

### **Control variables**

We selected 10 control variables related to respondents and CBIs. Regarding respondents, we controlled for age, gender, ethnicity (native vs. non-native Dutch), educational level, status of employment and organizational background. We also controlled for the role of the respondent in the initiative, asking respondents which role best describes their involvement in the initiative (board member (reference category), active member/volunteer (involved as a volunteer in a specific project), passive or supporting member/volunteer, and otherwise involved, recoded into missings). We specifically focused on their role concerning the management of the initiative, and not on their potential role as service user. Furthermore, we included the average number of hours (per week) the respondent participates in the initiative. Regarding the organization, we included the phase of the initiative as control, which has 5 categories: 1 = initial phase – reference category (researching, preparing, experimenting; acting as reference category), 2 = growing phase (mobilizing supporters, recognition by established parties), 3 = mature phase (fully operational), 4 = upscaling phase (exploring additional, new services), 5 = finishing phase (initiative is drawing to a close, completion). The last control variable we included is the sector in which the CBIs operate (16 sectors, ranging from healthcare, sustainable energy to education, and recreation).

### **Results**

Table 2 shows the means, standard deviations, and correlations for all model constructs and control variables. Regarding most core variables, respondents score them around the mid-range of the scales or slightly above this range, indicating (slightly) positive evaluations. Furthermore, the (significant) correlations with performance are positive for all factors and indicate that our expected relationships are likely to occur. See Appendix 2 for more information on descriptive results. Regarding performance, the average score for all seven dimensions is 4.60 on a seven-point scale. The scoring on social cohesion was highest ( $M = 5.10$ ,  $SD = 1.30$ ) and lowest on efficiency ( $M = 3.99$ ,  $SD = 1.62$ ). In Appendix 2 we show the descriptive information on all dimensional scores. In general, we can argue that according to the respondents the CBIs have a reasonable performance.

Table 2. Means, Standard Deviations, and Correlations among SEM-model constructs and control variables.

| Construct | M     | SD    | P | TFL | BSL   | GS    | OCHR  | OCRS  | BOSC  | LSC   | BRSC  | RA     | PG   | PM     | PU     | PF     |
|-----------|-------|-------|---|-----|-------|-------|-------|-------|-------|-------|-------|--------|------|--------|--------|--------|
| P         | 4.60  | 1.04  | 1 |     |       |       |       |       |       |       |       |        |      |        |        |        |
| TFL       | 4.73  | 1.36  | - | 1   | .55** | .44** | .24** | .11** | .14** | .17** | .22** | -.06   | -.03 | .13**  | .12**  | -.14** |
| BSL       | 4.12  | 1.47  | - | -   | 1     | .67** | .10*  | .23** | .24** | .24** | .25** | .05    | .01  | .04    | .12**  | -.15** |
| GS        | 3.39  | 1.53  | - | -   | -     | 1     | .43** | .08*  | .21** | .37** | .34** | -.03   | .06  | -.05   | .12**  | -.10*  |
| OCHR      | 2.20  | 1.45  | - | -   | -     | -     | 1     | .08*  | .15** | .39** | .20** | -.11** | .03  | -.03   | .05    | -.03   |
| OCRS      | 1.56  | 1.13  | - | -   | -     | -     | -     | 1     | .12** | .19** | .12** | .02    | .05  | .05    | .07    | -.12** |
| BOSC      | 2.65  | 1.03  | - | -   | -     | -     | -     | -     | 1     | .31** | .25** | .02    | -.05 | .04    | .12**  | -.12** |
| LSC       | 8.47  | 6.09  | - | -   | -     | -     | -     | -     | -     | 1     | .39** | .00    | -.01 | -.03   | .12**  | -.06   |
| BRSC      | 5.28  | 2.72  | - | -   | -     | -     | -     | -     | -     | -     | .44** | .04    | .09* | -.11** | .18**  | -.10** |
| RA        | 56.45 | 14.10 | - | -   | -     | -     | -     | -     | -     | -     | 1     | -.06   | .03  | -.05   | .15**  | -.10** |
| PG        | N/A   | N/A   | - | -   | -     | -     | -     | -     | -     | -     | -     | 1      | -.07 | .02    | .09*   | -.00   |
| PM        | N/A   | N/A   | - | -   | -     | -     | -     | -     | -     | -     | -     | -      | 1    | 1      | -.21** | -.20** |
| PU        | N/A   | N/A   | - | -   | -     | -     | -     | -     | -     | -     | -     | -      | -    | -      | 1      | -.33** |
| PF        | N/A   | N/A   | - | -   | -     | -     | -     | -     | -     | -     | -     | -      | -    | -      | -      | 1      |

\*  $p < .05$ . \*\*  $p < .01$  (2-tailed). N is in between 577 and 671 (pairwise deletion of missing values). Only constructs of the SEM model are shown, including significant control variables. The scale minimum and maximum are provided in abbreviations in the list of names hereafter. P = performance (1–7); TFL = transformational leadership (1–7); BSL = boundary spanning leadership (1–7); GS = government support (1–7); OCHR = organizational capacity, human resources (1–6); OCRS = organizational capacity, revenue sources (0–6); BOSC = bonding social capital (0–5); LSC = linking social capital (0–32); BRSC: bridging social capital (0–12); RA = respondent age (control variable) (19–88); PG = phase growing of the initiative (control variable); PM = phase mature of the initiative (control variable); PU = phase upscaling (control variable); PF = phase finishing of the initiative (control variable). The other control variables are not significant in the SEM-model and therefore excluded from this table.

## Hypothesis testing

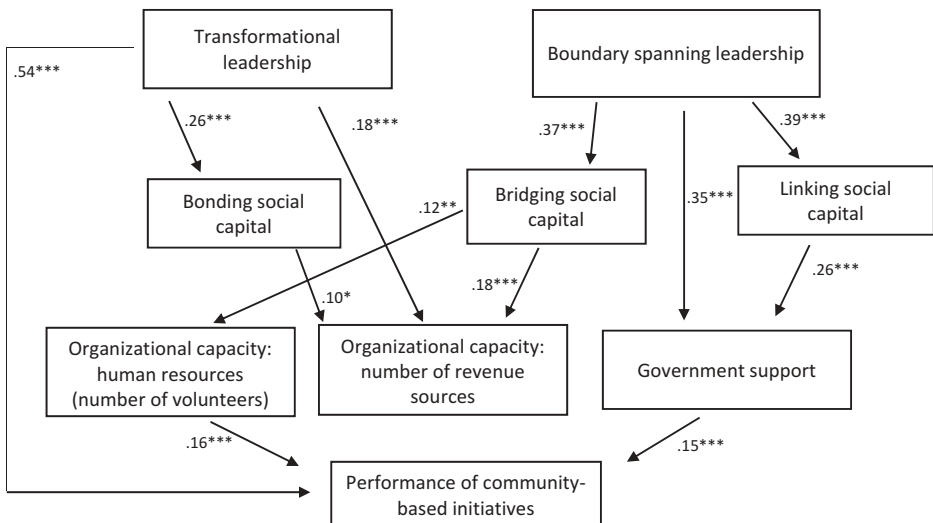
The overall fit of the structural model was tested using the following fit indices: CMIN/DF, the root-mean-square error of approximation (RMSEA), PCLOSE, and the comparative fit index (CFI).<sup>3</sup> Good model fit exists when CMIN/DF has a value between 1 and 3; RMSEA  $\leq 0.5$ ; PCLOSE  $\geq 0.5$ ; CFI  $\geq 0.95$  (Byrne 2010). However, in case of complex models (number of observed variables  $\geq 30$ ) and larger samples ( $N > 250$ ) less strict fit indices are common, with the value for CFI expected to be  $> .90$  and RMSEA  $<$  than .07 (Hair et al. 2010). Overall, the fit of the structural equation model to the data was good, with CMIN/DF: 2.64, RMSEA: .050, PCLOSE: .60; CFI: .94. Figure 2 presents the hypothesized results of this model.

## Direct impact of transformational leadership on performance

The results show that intra-organizational leadership has a direct relationship with performance of CBIs. Indeed, as expected in hypothesis 4, TFL has a strong and positive effect on performance. It turns out that more use of inspirational motivation, intellectual stimulation, and long-term visioning as community leadership style, increases the performance of CBIs.

## Impact of organizational capacity on performance

We distinguished among human resources and sources of revenue that can help initiatives to increase their performance. Hypothesis 1a.1, assumes a positive relationship between the number of volunteers and the performance of CBIs. Larger CBIs in terms of number of volunteers indeed show a higher level of performance.



**Figure 2.** Results of the structural equation model. Notes: \* $p \leq .05$ ; \*\* $p \leq .01$  \*\*\* $p \leq .001$ . The scores on all lines indicate the (direct) standardized regression coefficients (beta coefficients) for the significant relationships. The significant control variables are not depicted for display reasons. Function estimate means and intercepts used to deal with some missing values. The beta coefficients of the control variables (in relation to performance) are: age of respondent:  $-.10^{**}$ ; growing phase of initiative:  $.21^{**}$ ; mature phase of the initiative:  $.40^{***}$ ; upscaling phase of the initiative:  $.26^{***}$ ; finishing phase of initiative:  $.15^{**}$  (the phases have been correlated as well as the leadership styles).  $R^2$  Performance: .46;  $R^2$  Government support: .26;  $R^2$  Human resources: .02;  $R^2$  Revenue sources: .10;  $R^2$  Bonding social capital: .07;  $R^2$  Bridging social capital: .14;  $R^2$  Linking social capital: .15.

Surprisingly, however, having more diverse revenue sources are not found to be related with performance, and therefore hypothesis 1a.2 cannot be confirmed. Perhaps the presence of more revenue sources increases the burden upon citizens (e.g. having to meet requirements of various actors in exchange for financial support) in such an amount that it negates the benefits of having more revenue sources, for instance because of spending more time on paperwork instead of using the financial support to realize goals and increase social impact. The role of red tape is therefore interesting to examine in further research on performance of CBIs.

### ***Impact of government support on performance***

As [Figure 2](#) shows, the level of government support is positively related to performance of CBIs and is statistically significant, thereby confirming hypothesis 2. It seems that municipalities that assist CBIs in more than one way, using different types of support (e.g. supporting in obtaining extra resources and helping by providing availability to buildings) help increase performance of CBIs.

### ***The role of social capital***

[Figure 2](#) shows that bonding social capital has a significant and positive relationship with just one dimension of organizational capacity, the number of revenue sources, which means that hypothesis 3a.1 can be rejected and 3a.2 can be confirmed. Strong bonding ties among members of the core group are beneficial for the number of revenue sources a CBI can mobilize. Furthermore, [Figure 2](#) shows that linking social capital has a positive relationship with government support, thereby confirming hypothesis 3b. Having strong ties to linking actors, such as local government and funding agencies, increases the level of government support. In addition, in hypotheses 3c.1 and 3c.2 we expected that bridging social capital has positive relationships with organizational capacity. Both hypotheses are confirmed; strong bridging ties are positively related to human and financial resources of CBIs.

Furthermore, we hypothesized that the leadership styles have relationships with social capital. [Figure 2](#) shows that TFL has a positive relationship with bonding social capital, thereby confirming hypothesis 6. Transformational leadership seems to energize volunteers in the context of CBIs. Furthermore, BSL has positive relationships with both bridging and linking social capital, confirming hypotheses 7b.1 and 7b.2, showing the importance of this leadership style for mobilizing bridging and linking ties.

### ***Impact of transformational leadership on organizational capacity***

A positive relationship occurs between TFL and revenue sources. This result confirms hypothesis 5a.2 and rejects hypothesis 5a.1, as we expected that TFL positively influences both human and financial resources.

### ***Impact of boundary spanning leadership on government support***

[Figure 2](#) shows that BSL has a positive influence on government support, which confirms hypothesis 7a. This finding indicates that an inter-organizational leadership style helps to increase the level of government support that an initiative can obtain, which enhances the performance of the initiative (positive relationship between government support and performance). As government support is

an important resource for CBIs to mobilize, the finding indicates the relevance of boundary spanning activities and competences in acquiring the support.

### ***Relationships between the statistically significant control variables and performance***

The notes under [Figure 2](#) show the beta coefficients of the significant control variables. One individual characteristic is significant: the age of the respondent; older respondents tend to be more critical about performance. Furthermore, the evolution phases of CBIs are significant and positively related with performance.

## **Conclusions and discussion**

In this article, we have investigated which factors influence the performance of CBIs in the Netherlands.

Before we discuss our conclusions, we want to mention a few limitations of our study. First, we have to be careful in generalizing our results to other country contexts, as we have focused on CBIs in one country, the Netherlands. This country has specific governance and state traditions, which might explain the relationships between the variables. Different governance and state traditions may lead to different results (cf. Salamon and Anheier 1998). However, the CBIs in our sample share relevant characteristics, attributes that can be found in other contexts as well. Moreover, we strongly embedded our hypotheses in theory; literature that has been used in a wide variety of countries and contexts. Therefore, our results still have strong relevance for other CBI contexts. Second, our data are cross-sectional and causal inferences concerning the relationships in our structural model are based on theory. As is, in general, the case with survey research, a limitation lies in the possibility of reversed causality. Despite having strong theoretical imbedding of hypotheses, more research is needed to confirm our results. Moreover, we want to stress that in our presentation of results and conclusions, we do not intend causality, but relationships. Longitudinal and multiple source data on CBIs could provide more evidence on the feedback mechanisms between the factors impacting on performance and/or determine whether certain (combinations of) factors are (more) important in specific evolutionary phases of CBIs. Despite these limitations, we still think we can draw meaningful conclusions from our analysis.

On the performance of CBIs according to CBI participants, we can firstly conclude that CBIs manage to reach a satisfactory level. More research is needed on the question to what extent CBIs compensate for government retrenchment and/or lack of public policies in certain sectors. The results on perceived performance at least indicate that CBIs have potential impact on their local communities. Given more critical voices on the actual impact and scale of CBIs (e.g. Brandsen, Trommel, and Verschuere 2017), further debate and research are needed. Moreover, we adopted a subjective measure for performance for both theoretical and methodological reasons, but we acknowledge the limitation of this measure as a reliable proxy for objective performance – though both subjective and objective measures of performance have their own share of limitations, see for instance Andrews, Boyne, and Walker (2006) for a comparison of both measures in the public sector. Our research is among the first attempts to measure performance of CBIs in a systematic, quantitative (large N), and multi-categorical way. Important avenues for future research on

CBI's' performance, lie in the adoption of more tangible measures for performance (e.g. financial indicators, social impact measurement methods (see, for example, Maas and Liket (2011) for an overview of such methods), and in the inclusion of different viewpoints (e.g. surveying beneficiaries of CBIs or other relevant stakeholders, such as government officials) to capture the full picture of the construct.

Next to measuring performance, we tested an integrative theoretical model in which the performance of CBIs has been associated with key antecedents that show (distinctive yet interrelated) paths that explain performance. Three important paths that seem to enhance performance are a) by strong transformational leadership (TFL) as an intra-organizational style of community leadership, b) by boundary spanning leadership (BSL) as an inter-organizational leadership style that can mobilize bridging (communal) ties that increase organizational capacity in terms of human resources, and c) by BSL that can link CBIs with institutional partners, helping the CBIs in gaining support of government. Each path shows different and, until now, untested relationships between variables in the literature, and we therefore further in-depth discuss the main relationships.

The first relationship, and our second conclusion, we want to discuss is that different styles of community leadership are important to increase performance. Our research shows the importance of practising both leadership styles. Specifically, we showed that TFL has an important relationship with performance; it has by far the strongest association of all antecedents. TFL, and especially the dimension of intellectual stimulation, fits well with the nature of CBIs. CBIs are known to be innovative and driven by intrinsic motivations, trying to organize and operate public services in new ways and wanting to achieve higher-order societal goals (cf. Alford 2002; Voorberg, Bekkers, and Tummers 2015). This underlines the intellectual stimulating qualities of TFL that helps energizing and mobilizing a workforce that is eager to apply and develop its abilities on a job (cf. Hater and Bass 1988). Interestingly, even though the relevance of TFL has been debated by some scholars over the years (McCleskey 2014), this leadership style is obviously vital for these new forms of citizen-generated service delivery. Furthermore, TFL helps in realizing organizational capacity and social capital. Moreover, TFL rather than bonding social capital seems more important for mobilizing human resources, as no relationship was found between bonding ties and organizational capacity. Again, this shows how crucial inspirational leadership is for CBIs. Regarding organizational capacity, especially human resources have proven to be an important dimension to consider in research on performance of CBIs (cf. Foster-Fishman et al. 2001). Further research can analyse how citizens organize themselves, examining which institutional rules and design principles they initiate to create and maintain organizational capacity, taking a collective action perspective (cf. Ostrom 2000). In addition, the dimension of human resources could also be seen as an indicator for the size of the CBI. The perceived performance of CBIs turns out to be higher for larger CBIs; more volunteering hands on deck can help increase the (perceived) success of CBIs.

A third conclusion is that BSL, less rigorously examined as antecedent in previous research on outcomes of CBIs (cf. Igalla, Edelenbos, and Van Meerkerk *forthcoming*), is important to stimulate bridging and linking social capital and helps in realizing government support. Community boundary spanners are therefore important in building strategic alliances, mobilizing external resources and translating the initiative in such a way that both community and institutional actors understand its vision and goals and underline the relevancy of its work (Van Meerkerk and Edelenbos 2018).



Our fourth conclusion is that diverse forms of social capital can be utilized through their positive relationships with organizational capacity and government support, which both are positively associated with performance of CBIs. The underlying argument of this conclusion is based on (qualitative) research that argues to not only mobilize and create social capital, but also utilize the social ties in order to enhance outcomes of CBIs (cf. Dale and Newman 2010; Purdue 2001). Our research has tested this argument and we demonstrated that linking social capital is especially important for government support, that bridging social capital supports organizational capacity, and that, surprisingly, bonding social capital is not related to performance through this process of utilization.

Our fifth and final conclusion is about the role of government support. This article showed that government support is positively related to performance of CBIs. In line with previous (qualitative) research (e.g. Korosec and Berman 2006; Seixas and Berkes 2009), governmental institutions that express their support to CBIs in different ways, including allowance, counselling, and stimulation, can be instrumental to CBIs in realizing good performance. This is an interesting conclusion; however, the relationship with performance is not very strong, compared to for instance TFL. This observation requires further research; it is likely that (the strength of) government support can be crucial in certain phases of CBIs. In addition, as this antecedent does contribute to performance, it is important to know more about the reasons of governments to (not) support CBIs, and which characteristics of CBIs (e.g. evolution phases, size of CBI) are associated with higher levels of support. Moreover, it is interesting to test how different forms of support are related to performance; does variability in the type of government support also causes variation in performance of CBIs?

Our research reveals that CBIs can arrive at good outcomes when both intra- and inter-organizational styles of leadership are developed well, which relate to the quality of organizational capacity and the strength of government support, which in turn enhances performance. An important recommendation for future research, is to test the interrelationships between these antecedents in relation to performance, for instance by examining interaction effects. One possibility lies in testing whether an interaction between TFL as intra-organizational leadership style and BSL as inter-organizational leadership style has a stronger association with performance than either leadership style by itself.

To conclude, we believe that this article contributes to our knowledge of how important performance antecedents of CBIs relate to each other and to performance. We thereby make an effort to enhance the field of CBIs theoretically and methodologically.

## Notes

1. We performed PCA with oblique rotation in SPSS 24 and CFA analyses in Amos version 24 for all latent variables. Results showed clear one-component structures, meaning the scales for leadership styles and government support responded to the scales used in literature.
2. We performed within factor error correlations for two variables: performance (error correlation items 3 and 13, and 5 and 9) and government support (items 1 and 2, 2 and 3, 3 and 4, and 7 and 8).
3. See [Appendix 1](#) for the fit indices of the measurement model.



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## References

- Aldrich, H., and D. Herker. 1977. "Boundary Spanning Roles and Organization Structure." *Academy of Management Review* 2 (2): 217–230. doi:10.5465/amr.1977.4409044.
- Alford, J. 2002. "Why Do Public-Sector Clients Coproduce? toward a Contingency Theory." *Administration & Society* 34 (1): 32–56. doi:10.1177/0095399702034001004.
- Anderson, J., and D. Gerbing. 1988. "Structural Equation Modelling in Practice: A Review and Recommended Two-Step Approach." *Psychological Bulletin* 103: 411–423. doi:10.1037/0033-2909.103.3.411.
- Andrews, R., G. Boyne, and R. M. Walker. 2011. "The Impact of Management on Administrative and Survey Measures of Organizational Performance." *Public Management Review* 13 (2): 227–255. doi:10.1080/14719037.2010.532968.
- Andrews, R., G. A. Boyne, and R. M. Walker. 2006. "Subjective and Objective Measures of Organizational Performance: An Empirical Exploration." In *Public Service Performance:*

- Perspectives on Measurement and Management*, edited by Boyne, G., K. Meier, L. O'Toole, Jr., and R. Walker, 14–34. Cambridge: Cambridge University Press.
- Arnstein, S. R. 1969. "A Ladder of Citizen Participation." *Journal of the American Institute of Planners* 35 (4): 216–224. doi:10.1080/01944366908977225.
- Bagnoli, L., and C. Megali. 2011. "Measuring Performance in Social Enterprises." *Nonprofit and Voluntary Sector Quarterly* 40 (1): 149–165. doi:10.1177/0899764009351111.
- Bailey, N. 2012. "The Role, Organisation and Contribution of Community Enterprise to Urban Regeneration Policy in the UK." *Progress in Planning* 77 (1): 1–35. doi:10.1016/j.progress.2011.11.001.
- Bass, B. M., B. J. Avolio, D. I. Jung, and Y. Berson. 2003. "Predicting Unit Performance by Assessing Transformational and Transactional Leadership." *Journal of Applied Psychology* 88 (2): 207. doi:10.1037/0021-9010.88.2.207.
- Boonstra, B., and L. Boelens. 2011. "Self-Organization in Urban Development: Towards a New Perspective on Spatial Planning." *Urban Research & Practice* 4 (2): 99–122. doi:10.1080/17535069.2011.579767.
- Boyne, G. A. 2002. "Theme: Local Government: Concepts and Indicators of Local Authority Performance: An Evaluation of the Statutory Frameworks in England and Wales." *Public Money and Management* 22 (2): 17–24. doi:10.1111/pmam.2002.22.issue-2.
- Branden, T., W. Trommel, and B. Verschuere. 2017. "The State and the Reconstruction of Civil Society." *International Review of Administrative Sciences* 83 (4): 676–693. doi:10.1177/0020852315592467.
- Brown, T. A. 2015. *Confirmatory Factor Analysis for Applied Research*. 2nd ed. New York: Guilford Publications.
- Byrne, B. M. 2010. *Structural Equation Modeling with AMOS. Basic Concepts, Applications, and Programming*. New York: Routledge.
- Conway, J. M., and C. E. Lance. 2010. "What Reviewers Should Expect from Authors regarding Common Method Bias in Organizational Research." *Journal of Business and Psychology* 25 (3): 325–334. doi:10.1007/s10869-010-9181-6.
- Dale, A., and L. Newman. 2010. "Social Capital: A Necessary and Sufficient Condition for Sustainable Community Development?" *Community Development Journal* 45 (1): 5–21. doi:10.1093/cdj/bsn028.
- Dietz, T., E. Ostrom, and P. C. Stern. 2003. "The Struggle to Govern the Commons." *Science* 302 (5652): 1907–1912. doi:10.1126/science.1091015.
- Douglas, M. 1991. "The Idea of A Home: A Kind of Space." *Social Research* 58 (1): 287–307.
- Edelenbos, J., and I. Van Meerkerk, Eds. 2016. *Critical Reflections on Interactive Governance: Self-Organization and Participation in Public Governance*. Cheltenham: Edward Elgar Publishing.
- Edelenbos, J., I. Van Meerkerk, and T. Schenk. 2018. "The Evolution of Community Self-Organization in Interaction with Government Institutions Cross-Case Insights from Three Countries." *The American Review of Public Administration* 48 (1): 52–66. doi:10.1177/0275074016651142.
- Eisinger, P. 2002. "Organizational Capacity and Organizational Effectiveness among Street-Level Food Assistance Programs." *Nonprofit and Voluntary Sector Quarterly* 31 (1): 115–130. doi:10.1177/0899764002311005.
- Foster-Fishman, P. G., S. L. Berkowitz, D. W. Lounsburt, S. Jacobson, and N. A. Allen. 2001. "Building Collaborative Capacity in Community Coalitions: A Review and Integrative Framework." *American Journal of Community Psychology* 29 (2): 241–261. doi:10.1023/A:1010378613583.
- George, B., and S. K. Pandey. 2017. "We Know the Yin—But Where Is the Yang? toward a Balanced Approach on Common Source Bias in Public Administration Scholarship." *Review of Public Personnel Administration* 37 (2): 245–270. doi:10.1177/0734371X17698189.
- Hair, J. F., W. C. Black, B. J. Babin, and R. E. Anderson. 2010. *Multivariate Data Analysis: A Global Perspective*. 7th ed. New Jersey: Pearson Prentice Hall.
- Hartley, J., and M. Allison. 2000. "The Modernization and Improvement of Government and Public Services: The Role of Leadership in the Modernization and Improvement of Public Services." *Public Money and Management* 20 (2): 35–40. doi:10.1111/1467-9302.00209.

- Hater, J. J., and B. M. Bass. 1988. "Superiors' Evaluations and Subordinates' Perceptions of Transformational and Transactional Leadership." *Journal of Applied Psychology* 73 (4): 695. doi:10.1037/0021-9010.73.4.695.
- Healey, P. 2015. "Citizen-Generated Local Development Initiative: Recent English Experience." *International Journal of Urban Sciences* 19 (2): 109–118. doi:10.1080/12265934.2014.989892.
- Igalla, M., J. Edelenbos, and I. Van Meerkerk. *forthcoming*. "Citizens in Action, What Do They Accomplish? A Systematic Literature Review of Citizen Initiatives, Their Main Characteristics, Outcomes, and Factors."
- Kendall, J., and M. Knapp. 2000. "Measuring the Performance of Voluntary Organizations." *Public Management Review* 2 (1): 105–132. doi:10.1080/14719030000000006.
- Klijn, E.-H., J. Edelenbos, and B. Steijn. 2010. "Trust in Governance Networks: Its Impacts on Outcomes." *Administration & Society* 42 (2): 193–221. doi:10.1177/0095399710362716.
- Korosec, R. L., and E. M. Berman. 2006. "Municipal Support for Social Entrepreneurship." *Public Administration Review* 66 (3): 448–462. doi:10.1111/puar.2006.66.issue-3.
- Liu, G., S. Takeda, and -W.-W. Ko. 2014. "Strategic Orientation and Social Enterprise Performance." *Nonprofit and Voluntary Sector Quarterly* 43 (3): 480–501. doi:10.1177/0899764012468629.
- Llano-Arias, V. 2015. "Community Knowledge Sharing and Co-production of Water Services: Two Cases of Community Aqueduct Associations in Colombia." *Water Alternatives* 8 (2): 77–98.
- Maas, K., and K. Liket. 2011. "Social Impact Measurement: Classification of Methods." In *Environmental Management Accounting and Supply Chain Management*, edited by Burritt, R. L., S. Schaltegger, M. Bennett, T. Pohjola and M. Csutora, 171–202. Dordrecht: Springer.
- McCleskey, J. A. 2014. "Situational, Transformational, and Transactional Leadership and Leadership Development." *Journal of Business Studies Quarterly* 5 (4): 117.
- Miller, P. M. 2008. "Examining the Work of Boundary Spanning Leaders in Community Contexts." *International Journal of Leadership in Education* 11 (4): 353–377. doi:10.1080/13603120802317875.
- Nabatchi, T. 2012. "Putting the 'Public' Back in Public Values Research: Designing Participation to Identify and Respond to Values." *Public Administration Review* 72 (5): 699–708. doi:10.1111/puar.2012.72.issue-5.
- Newman, L., L. Waldron, A. Dale, and K. Carriere. 2008. "Sustainable Urban Community Development from the Grassroots: Challenges and Opportunities in a Pedestrian Street Initiative." *Local Environment* 13 (2): 129–139. doi:10.1080/13549830701581879.
- Nicholls, W. 2009. "Place, Networks, Space: Theorising the Geographies of Social Movements." *Transactions of the Institute of British Geographers* 34 (1): 78–93. doi:10.1111/tran.2009.34.issue-1.
- Osborne, S. P., Z. Radnor, and K. Strokosch. 2016. "Co-Production and the Co-Creation of Value in Public Services: A Suitable Case for Treatment?" *Public Management Review* 18 (5): 639–653. doi:10.1080/14719037.2015.1111927.
- Ostrom, E. 2000. "Collective Action and the Evolution of Social Norms." *Journal of Economic Perspectives* 14 (3): 137–158. doi:10.1257/jep.14.3.137.
- Phillips, R., and R. Pittman, eds. 2009. *An Introduction to Community Development*. London: Routledge.
- Podsakoff, P. M., S. B. MacKenzie, J. Y. Lee, and N. P. Podsakoff. 2003. "Common Method Biases in Behavioral Research: A Critical Review of the Literature and Recommended Remedies." *Journal of Applied Psychology* 88: 879–903. doi:10.1037/0021-9010.88.5.879.
- Purdue, D. 2001. "Neighbourhood Governance: Leadership, Trust and Social Capital." *Urban Studies* 38 (12): 2211–2224. doi:10.1080/00420980120087135.
- Putnam, R. D. 1995. "Tuning In, Tuning Out: The Strange Disappearance of Social Capital in America." *PS: Political Science & Politics* 28 (4): 664–684.
- Roberts, N. 2004. "Public Deliberation in an Age of Direct Citizen Participation." *The American Review of Public Administration* 34 (4): 315–353. doi:10.1177/0275074004269288.
- Salamon, L. M., and H. K. Anheier. 1998. "Social Origins of Civil Society: Explaining the Nonprofit Sector Cross-Nationally." *Voluntas: International Journal of Voluntary and Nonprofit Organizations* 9 (3): 213–248. doi:10.1023/A:1022058200985.
- Seixas, C., and F. Berkes. 2009. "Community-Based Enterprises: The Significance of Partnerships and Institutional Linkages." *International Journal of the Commons* 4 (1): 183. doi:10.18352/ijc.133.
- Sharir, M., and M. Lerner. 2006. "Gauging the Success of Social Ventures Initiated by Individual Social Entrepreneurs." *Journal of World Business* 41 (1): 6–20. doi:10.1016/j.jwb.2005.09.004.

- Szreter, S. 2002. "The State of Social Capital: Bringing Back in Power, Politics, and History." *Theory and Society* 31 (5): 573–621. doi:[10.1023/A:1021300217590](https://doi.org/10.1023/A:1021300217590).
- Szreter, S., and M. Woolcock. 2004. "Health by Association? Social Capital, Social Theory, and the Political Economy of Public Health." *International Journal of Epidemiology* 33: 650–667. doi:[10.1093/ije/dyh013](https://doi.org/10.1093/ije/dyh013).
- Torring, J., E. Sørensen, and A. Røiseland. 2019. "Transforming the Public Sector into an Arena for Co-Creation: Barriers, Drivers, Benefits, and Ways Forward." *Administration & Society* 51 (5): 1–31. doi: [10.1177/0095399716680057](https://doi.org/10.1177/0095399716680057)
- Van Meerkerk, I., and J. Edelenbos. 2018. *Boundary Spanners in Public Management and Governance: An Interdisciplinary Assessment*. Cheltenham: Edward Elgar Publishing.
- Varda, D. M. 2011. "A Network Perspective on State–Society Synergy to Increase Community- Level Social Capital." *Nonprofit and Voluntary Sector Quarterly* 40 (5): 1–28. doi: [10.1177/0899764010378187](https://doi.org/10.1177/0899764010378187)
- Voorberg, W. H., V. J. J. M. Bekkers, and L. G. Tummers. 2015. "A Systematic Review of Co-Creation and Co-Production: Embarking on the Social Innovation Journey." *Public Management Review* 17 (9): 1333–1357. doi:[10.1080/14719037.2014.930505](https://doi.org/10.1080/14719037.2014.930505).
- Williams, P. 2002. "The Competent Boundary Spanner." *Public Administration* 80 (1): 103–124. doi:[10.1111/padm.2002.80.issue-1](https://doi.org/10.1111/padm.2002.80.issue-1).
- Wright, B. E., D. P. Moynihan, and S. J. Pandey. 2012. "Pulling the Levers: Transformational Leadership, Public Service Motivation, and Mission Valence." *Public Administration Review* 72 (2): 206–215. doi:[10.1111/j.1540-6210.2011.02496.x](https://doi.org/10.1111/j.1540-6210.2011.02496.x).

Appendix 1. Further information on methodology

Scale validation performance

Before we performed the SEM analyses, we validated the instrument for performance. To do this, we randomly split our sample into two subsamples. We used subsample A (N = 335) to explore the component structure of the instrument using PCA with oblique rotation in SPSS 23. We used subsample B (N = 336) for cross-validation of the instrument using confirmatory factor analysis in AMOS 22, and for the measurement model. The entire sample (N = 671) was then used for an extra validation check for performance using CFA, as well as for the measurement model, and structural model (see Table A1). The fit indices of the measurement model of both subsample B and the main sample show a good fit of the model. For subsample B the fit indices are: CMIN/DF: 1.75; RMSEA: .047; PCLOSE: .75; CFI: .97. For the main sample, the fit indices are: CMIN/DF: 2.63; RMSEA: .049; PCLOSE: .60; CFI: .96.

To develop a scale for performance, we started with 14 items that cover the above mentioned seven categories/dimensions. The initial PCA analyses (with subsample A) showed the existence of multiple components, however, these were not relevant theoretically. The fact that we measured each dimension with two rather than three items will have had a role in this observation. Afterwards, we decided to measure performance as a first-order construct instead of a second-order variable. We also concluded that two items did not capture the content of the categories after all, which meant that we needed to exclude one item for each of the other five categories as well, in order to prevent dominance of certain performance criteria. Based on theoretical relevance and correlations, we excluded one item of each category. Thereafter, we performed a PCA (see Table A1), which showed a clear one-component structure according to the proposed categories. We used CFA to further test the PCA structure on subsample B. The standardized factor loadings (see Table A1) are in a reasonable range (all greater than the minimum of .4 and, except for one item, all items score above .62). The measurement model indicates (for performance) standardized factor loadings that are  $\geq .51$  and range between .51 and .78 (see Table A2) which indicates reasonable to good convergent validity (Hair et al. 2010). Furthermore, the composite reliability (see Table A2) is high (.85) and exceeds the .70 threshold. In addition, the Cronbach's Alpha (see Table A2) is greater than the threshold of .80. The average variance extracted (AVE) is slightly lower (.44) than the generally acceptable value of .5 (see Table A2), which indicates a limitation for the convergent validity. On the contrary, the AVE is larger than the squared inter-construct correlations of all four constructs, indicating the distinctiveness of the construct and, thus, discriminant validity (see Table A2).

The reliability and validity of the constructs in the measurement model of subsample B is as follows: all factor loadings are  $\geq .51$  and most go beyond .70; AVE of one construct is lower than .50 (performance); all AVEs are higher than their corresponding squared correlations; all Cronbach's Alpha's are higher than .80; all composite reliability values are higher than .80.

**Table A1.** Results of the performance scale validation on subsample A (PCA; N = 335), B (CFA; N = 336), and main sample (CFA; N = 671).

|                                                | Item 1 | Item 2 | Item 3 | Item 4 | Item 5 | Item 6 | Item 7 |
|------------------------------------------------|--------|--------|--------|--------|--------|--------|--------|
| PCA component loadings                         | .80    | .74    | .75    | .75    | .81    | .77    | .59    |
| CFA standardized factor loadings (subsample B) | .77    | .63    | .65    | .75    | .74    | .64    | .49    |
| CFA standardized factor loadings (main sample) | .77    | .66    | .69    | .72    | .77    | .69    | .50    |

Notes: for CFA, all factor loadings  $p < .001$ . PCA and CFA were performed for performance only. Subsample A was used for PCA analyses; subsample B was used for cross-validation using CFA analyses (and the measurement model); the main sample was used for SEM analyses (measurement model and structural model).

**Table A2.** Measurement items and construct's reliability subsample B (N = 336).

| Constructs and items                                                                                     | Standardized factor loadings | AVE (convergent validity)/squared correlation (discriminant validity) | Cronbach's Alpha/composite reliability |
|----------------------------------------------------------------------------------------------------------|------------------------------|-----------------------------------------------------------------------|----------------------------------------|
| <b>Community-based initiative performance</b>                                                            |                              |                                                                       |                                        |
| The citizen initiative in which I am (was) involved ...                                                  |                              |                                                                       |                                        |
| 1) achieves its objectives                                                                               | .78                          | 0.44/                                                                 | 0.84/0.85                              |
| 2) provides answers to key issues                                                                        | .61                          | Performance – TFL: 0.37                                               |                                        |
| 3) creates better connections between residents/citizens (social cohesion)                               | .63                          | Performance – BSL: 0.21                                               |                                        |
| 4) manages to deliver high-quality services and goods                                                    | .76                          | 0.21                                                                  |                                        |
| 5) is considered important by the community                                                              | .70                          | Performance – GS: 0.08                                                |                                        |
| 6) finds smarter solutions to problems than previous attempts                                            | .61                          |                                                                       |                                        |
| 7) earns a sufficient income to cover expenses                                                           | .51                          |                                                                       |                                        |
| <b>Transformational leadership (TFL)</b>                                                                 |                              |                                                                       |                                        |
| The management (board or leading persons) of the initiative ...                                          |                              |                                                                       |                                        |
| 1) has a clear idea (vision) of where the organization is heading                                        | .81                          | 0.68/                                                                 | .91/.91                                |
| 2) is a source of inspiration for the members/volunteers within its organization                         | .88                          | TFL – BSL: 0.59                                                       |                                        |
| 3) makes plans that display visible leadership to its members/volunteers and followers                   | .85                          | TFL – GS: 0.09                                                        |                                        |
| 4) develops long-term plans for the organization                                                         | .74                          |                                                                       |                                        |
| 5) challenges me to solve existing problems in new ways                                                  | .83                          |                                                                       |                                        |
| <b>Boundary spanning leadership (BSL)</b>                                                                |                              |                                                                       |                                        |
| The management (board or leading persons) of the initiative ...                                          |                              |                                                                       |                                        |
| 1) involves people from outside the organization when making decisions                                   | .80                          | 0.70/                                                                 | .92/.92                                |
| 2) works closely with others (outside the organization) to achieve results                               | .85                          | BSL – GS: 0.22                                                        |                                        |
| 3) aims to link external developments (new opportunities, possibilities, etc.) to the citizen initiative | .88                          |                                                                       |                                        |
| 4) devotes a lot of time to maintaining contact with parties outside the organization                    | .84                          |                                                                       |                                        |
| 5) has a sense of what is important for parties outside the initiative                                   | .83                          |                                                                       |                                        |

(Continued)

Table A2. (Continued).

| Constructs and items                                                                                                             | Standardized factor loadings | AVE (convergent validity)/squared correlation (discriminant validity) | Cronbach's Alpha/composite reliability |
|----------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----------------------------------------------------------------------|----------------------------------------|
| <b>Government support (GS)</b>                                                                                                   |                              |                                                                       |                                        |
| The municipality ...                                                                                                             |                              | 0.69                                                                  | .96/.96                                |
| 1) provides financial assistance to the initiative (such as subsidies)                                                           | .72                          |                                                                       |                                        |
| 2) supports the initiative in obtaining extra resources (e.g. acquisition, submitting applications for subsidies, fundraising)   | .81                          |                                                                       |                                        |
| 3) helps the initiative by providing availability to real estate (buildings) or land                                             | .64                          |                                                                       |                                        |
| 4) gives the initiative the opportunity to execute assignments (e.g. waste collection, maintenance of public green spaces, etc.) | .76                          |                                                                       |                                        |
| 5) supports the initiative through the provision of information                                                                  | .88                          |                                                                       |                                        |
| 6) contributes towards initiative awareness                                                                                      | .89                          |                                                                       |                                        |
| 7) assists in the coordination with other involved parties                                                                       | .90                          |                                                                       |                                        |
| 8) encourages collaboration between those parties with an interest in the initiative                                             | .89                          |                                                                       |                                        |
| 9) provides the initiative with advice, if required                                                                              | .89                          |                                                                       |                                        |
| 10) actively cooperates with the initiative                                                                                      | .90                          |                                                                       |                                        |
| 11) participates in the initiative by taking responsibility for certain tasks                                                    | .80                          |                                                                       |                                        |
| <b>Organizational capacity</b>                                                                                                   |                              |                                                                       |                                        |
| Revenue sources                                                                                                                  | N/A                          | N/A                                                                   | N/A                                    |
| How does the citizen initiative earn an income? Multiple answers are possible:                                                   |                              |                                                                       |                                        |
| 1) subsidies from the local government (e.g. the municipality) (yes/no)                                                          |                              |                                                                       |                                        |
| 2) subsidies from the regional government (e.g. county/province or city region) (yes/no)                                         |                              |                                                                       |                                        |
| 3) subsidies from the Dutch government (e.g. ministries) (yes/no)                                                                |                              |                                                                       |                                        |
| 4) services to the government (e.g. via a contract or social contract) (yes/no)                                                  |                              |                                                                       |                                        |
| 5) private revenue (the sale of goods and services to a non-governmental party) (yes/no)                                         |                              |                                                                       |                                        |
| 6) donations (yes/no)                                                                                                            |                              |                                                                       |                                        |
| 7) fundraising (e.g. by a foundation or bank) (yes/no)                                                                           |                              |                                                                       |                                        |
| 8) contributions (from members/volunteers) (yes/no)                                                                              |                              |                                                                       |                                        |
| Human resources                                                                                                                  |                              |                                                                       |                                        |
| How many people (volunteers and/or active members) are active in the citizen initiative (scale 1-6)?                             |                              |                                                                       |                                        |

(Continued)

Table A2. (Continued).

| Constructs and items                                                                                                                                                                                                                     | Standardized factor loadings | AVE (convergent validity)/squared correlation (discriminant validity) | Cronbach's Alpha/composite reliability |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|-----------------------------------------------------------------------|----------------------------------------|
| <b>Social capital</b>                                                                                                                                                                                                                    |                              |                                                                       |                                        |
| <u>Bonding social capital</u>                                                                                                                                                                                                            |                              |                                                                       |                                        |
| - How frequently do members of the core group interact with issues that affect the citizen initiative? This interaction can be in various forms, such as attending management meetings, organizing activities and/or providing services. | N/A                          | N/A                                                                   | N/A                                    |
| <u>Bridging and linking social capital</u>                                                                                                                                                                                               |                              |                                                                       |                                        |
| Generally speaking, how often is the citizen initiative in contact with the following parties? This contact can take various forms, e.g. working together, performing a financial consultation and/or receiving advice.                  |                              |                                                                       |                                        |
| Linking:                                                                                                                                                                                                                                 |                              |                                                                       |                                        |
| Central government, province (county), municipality: local council, municipality: municipal executive, municipality: civil servants, funds/sponsors, housing corporations.                                                               |                              |                                                                       |                                        |
| Bridging:                                                                                                                                                                                                                                |                              |                                                                       |                                        |
| Residents organizations/other community initiatives, and visitors or users of the initiative                                                                                                                                             |                              |                                                                       |                                        |

Note: N = 336. All factor loadings:  $p < .001$



## Appendix 2. More information on descriptive results

The data show a lot of variation in the sectors in which the community-based initiatives tackle problems. Examples of these sectors (or objectives) are enhancing well-being (37.11%), strengthening social cohesion (38.00%), and enhancing livability (50.37%). Most initiatives are focused at enhancing livability in their neighbourhood, whereas increasing entrepreneurship seems to be the least popular objective among the initiatives (3.13%).

Regarding the status of development or evolutionary phase of the initiatives, we see that most of the initiatives (47.24%) are in a mature phase (fully operational), 24.44% is in a growth phase, indicating a developing status in which support is being mobilized as well as recognition by established parties. The phase of upscaling (11.92%) is the third most common phase, referring to activities like exploring new services. In addition, 10.73% of the initiatives can be found in a finishing phase, meaning the initiative has ended. Finally, 5.66% of the initiatives is in an initial phase, indicating that they are active in activities like researching, preparing, and experimenting.

When comparing the average scores for the different items of performance, we see that the initiatives have scored item three (the citizen initiative creates better connections between residents/citizens – social cohesion) the highest ( $M = 5.10$ ;  $SD = 1.30$ ), indicating the relative importance of social impact of citizen initiatives. Goal effectiveness (do the initiatives deliver what they were designed to deliver) is the item with the second highest average score ( $M = 4.92$ ;  $SD = 1.31$ ), showing slightly positive scores ranging between a 4 and a 6 on a seven-point scale. With an average score of 4.89 ( $SD = 1.39$ ), legitimacy (being considered as important by the community) has also been evaluated as slightly positive, indicating that community members value the services provided by citizen initiatives. Regarding to the quality of services, an average score of 4.77 ( $SD = 1.37$ ) has been given by the respondents, showing the same overall slightly positive score meaning that the respondents consider themselves to be able to deliver high-quality services and goods for their community. Interestingly, problem-solving capacity ( $M = 4.21$ ;  $SD = 1.50$ ), innovativeness ( $M = 4.32$ ;  $SD = 1.50$ ), and efficiency ( $M = 3.99$ ;  $SD = 1.62$ ) have been scored less positive compared to social impact, legitimacy, and quality of services. It seems that the community initiatives are less concerned with finding smarter solutions to problems with their services than previous attempts, addressing key problems with their services, and earning enough income to cover the expenses.